

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 13

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Lab Assignment: 13

Aim:

Write a program to calculate the value of an investment. Consider an Initial value and annual interest as an input and calculate the value of the investment over time.

Objective:

- To understand the concept of investment and interest
- To use loop for repeated calculation
- To calculate investment value over time

Theory:

Investment value increases over time due to interest.

The annual interest rate is applied every year to the current investment value.

Using a loop, the investment value is updated year by year and displayed.

Formula used:

$$\text{Investment Value} = \text{Investment Value} + (\text{Investment Value} \times \text{Interest Rate} / 100)$$

Algorithm:

- Start
- Read initial investment amount
- Read annual interest rate
- Read number of years
- Set investment value equal to initial amount
- Repeat for each year
 - Calculate interest
 - Add interest to investment value
 - Round the value up to 3 decimals
 - Display the value
- Stop

Pseudo Code:

START

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```
READ amount
READ rate
READ years
value = amount
FOR i = 1 TO years
    value = value + (value * rate / 100)
    value = round(value, 3)
    PRINT value
END FOR
STOP
```

Code:

```
amount = float(input("Enter initial investment amount: "));
rate = float(input("Enter annual interest rate: "));
years = int(input("Enter number of years: "));
value = amount;
for i in range(1, years + 1):
    value = value + (value * rate / 100);
    print("Value at year", i, "is:", round(value, 3));
```

Output:

```
Enter initial investment amount: 10000
Enter annual interest rate: 18
Enter number of years: 10
Value at year 1 is: 11800.0
Value at year 2 is: 13924.0
Value at year 3 is: 16430.32
Value at year 4 is: 19387.778
Value at year 5 is: 22877.578
Value at year 6 is: 26995.542
Value at year 7 is: 31854.739
Value at year 8 is: 37588.592
Value at year 9 is: 44354.539
Value at year 10 is: 52338.356
```

Conclusion:

The program successfully calculates the investment value over time and displays the result rounded up to three decimal places using loops and arithmetic operations.